

Project ID:

25-26J-067

1. Topic (12 words max)

“SafeLAN”: Adaptive Secure File System for LAN environment

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Despite extensive research in the domain of secure file systems, there remains a significant gap in the development of security solutions specifically designed for Local Area Networks (LANs) that are independent of internet connectivity or default operating system-level permissions. Most of the existing secure file systems and access control mechanisms depend heavily on either cloud-based services or some form of online infrastructure, making them infeasible for an organization that works in a LAN-only environment, where there are restrictions on internet policies, concerns about latencies, or simply requirements for data confidentiality. Other typical ways of protecting files, like using OS-level permissions, static passwords, or simple encryption do not have the capability for ongoing authentication and cannot identify unusual user actions.

This research identifies specific problems in the following areas:

1. Traditional ways of authentication do not validate access dynamically in real-time or do not adapt to user behaviour. Therefore, insider threats and illegal data access cannot be detected once the user has logged in. Existing LAN environments lack file systems that offer real-time, dynamic identity verification based on behavioral characteristics.
2. In today's organizations, threats often occur after login, during active sessions, due to unattended workstations, remote takeovers, or MITM attacks. Traditional authentication cannot verify user identity continuously, exposing systems in hospitals, banks, and government offices to misuse. A smart, non-intrusive machine learning-based solution is needed to monitor user behavior in real time, detect anomalies, and prevent hijacking during active sessions.
3. For the third component, we develop a solution considering privacy when it comes to face recognition. Some users would worry about how their face identity will be stored and used in systems. And develop a more secure file transfer protocol less prone to brute force attacks that current available protocols utilizing quantum cryptography.
4. Conventional file access logs in LANs are stored locally and can be easily altered or deleted by administrators or attackers. do not have transparency, immutability, or responsibility. Most Small to Medium Enterprises (SMEs) do not deploy centralized and tamper-evident audit trails, hence making post-incident investigations very challenging.

Given these limitations, the research need is to design a LAN-focused secure file system solution that has advanced authentication, behavioral intelligence, secure file access, and auditable traceability without depending on the internet.

References:

[1] K. Aditya, Sannuthi Krishna Nihar, and V. S. Saranya, "Advanced Authentication and Secure File Management System with Multi-Modal Security Integration and Role-Based Access Control," *SSRN Electronic Journal*, Jan. 2025, doi: <https://doi.org/10.2139/ssrn.5089188>.

[2] "Google Scholar," *Google.com*, 2025.

https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=behaviour+anomaly+detection+using+machine+learning+to+ensure+continuous+authentication&btnG=#d=gs_qabs&t=1750907486144&u=%23p%3DY-IOA-ufKiOJ (accessed Jun. 27, 2025).

[3] Payam Almasian and Alireza Shafieinejad, "Secure cloud file sharing scheme using blockchain and attribute-based encryption," *Computer Standards & Interfaces*, vol. 87, pp. 103745–103745, Jan. 2024, doi: <https://doi.org/10.1016/j.csi.2023.103745>.

[4] K. W. Bowyer, "Face recognition technology: security versus privacy," *IEEE Technology and Society Magazine*, vol. 23, no. 1, pp. 9–19, 2004, doi: <https://doi.org/10.1109/mtas.2004.1273467>.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a secure desktop-based application named "SafeLAN" for LAN environments, offering a comprehensive framework for secure file sharing, continuous authentication, intelligent monitoring, and transparent auditing, without requiring internet access. This solution is especially made for small to medium businesses (SMEs) that value and prioritize usability, cost-efficiency, and data privacy.

The system enhances traditional file-sharing by introducing a multi-layered security architecture. Upon user login, the application uses behavioural biometrics such as keystroke dynamics and mouse movements to keep checking that the right user is still there during the session. This real-time dynamic access control helps prevent insider threats.

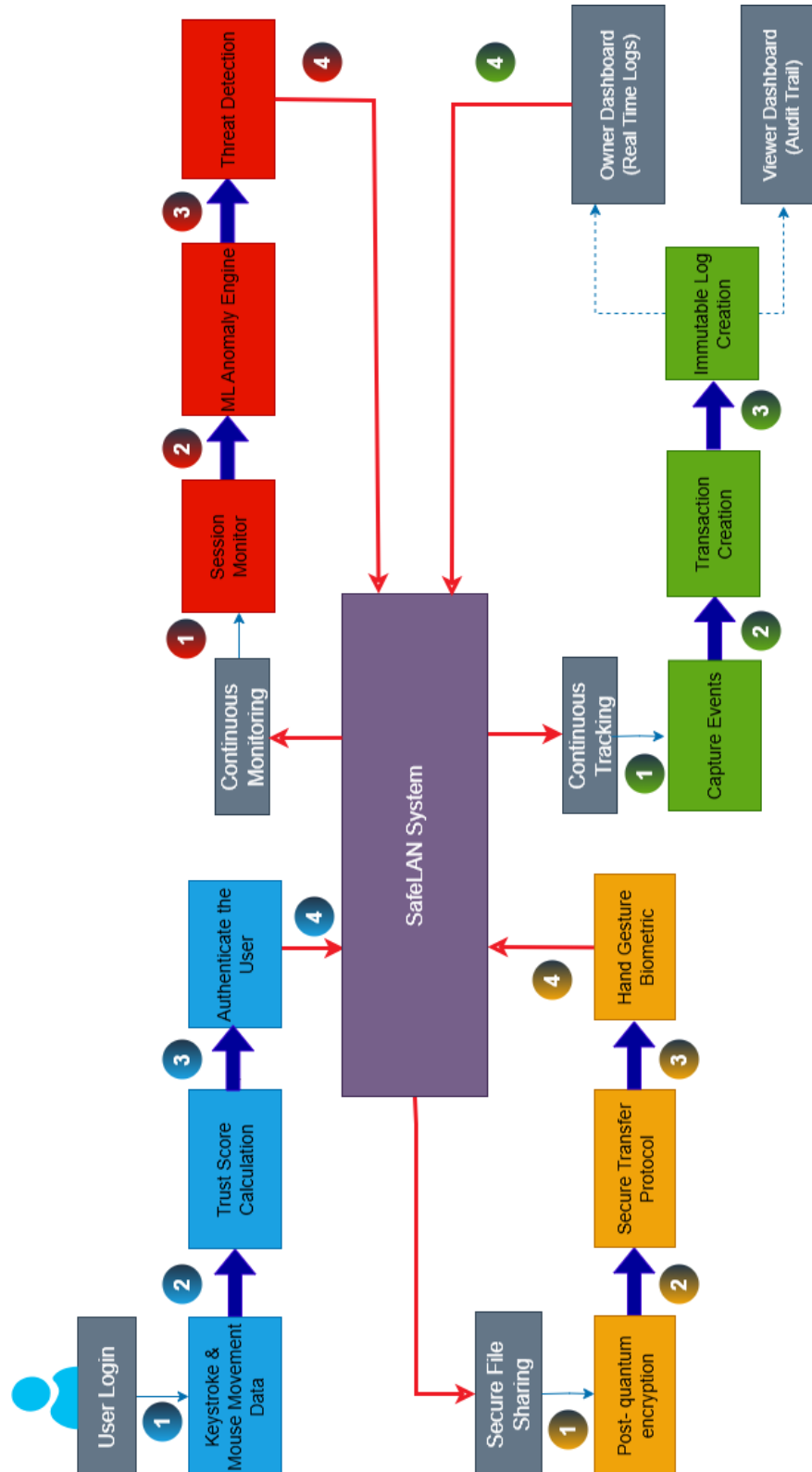
In order to authenticate the user continuously, machine learning helps in detecting behavioral anomalies in the LAN environment. After the user logged into the system, it monitors user actions like typing and mouse patterns to identify threats such as physical hijacking, remote session hijacking, and MITM attacks, ensuring real-time protection by locking sessions or alerting administrators when anomalies occur.

Secure file viewer helps to ensure protected access to sensitive files using post-quantum encryption, gesture-based OTP authentication, and a lockdown environment. By ensuring the confidentiality and integrity within the LAN environment, it helps in preventing unauthorized actions like accessing sensitive files and modifying them, by ensuring.

Finally, every access event is recorded using a permissioned blockchain-based audit logging system, which guarantees tamper-evident, immutable records of file activity. This ensures accountability and simplifies incident investigation.

"SafeLAN" application has been built with an intuitive Graphical User Interface (GUI) that is accessible to non-technical users. Completely offline operation within the LAN makes this application a practical, robust solution for secure, monitored, controlled file access within enterprises.

Conceptual Diagram



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The proposed research lies at the intersection of cybersecurity, machine learning, blockchain, and human-computer interaction, specifically within the domain of secure LAN-based systems. Therefore, the project demands specialized domain expertise and knowledge along with data requirements for the ML components.

Domain expertise and knowledge:

- **Information and Network Security:** Expertise in encryption techniques such as Advanced Encryption Standard (AES), secure file transmission protocols, and local file system security is crucial to implement a robust and independent file sharing system within a LAN environment. The ability to secure data without relying on internet-based services is a fundamental requirement of the solution.
- **Behavioural Biometrics:** Knowledge to work with keystroke and mouse movement data is necessary to ensure the dynamic behavioural authentication. For that expertise in Machine Learning (ML) or Artificial Intelligence is needed (AI). A trust scoring mechanism also need to be implemented to securely authenticate the user with the help of user behaviour profile and other contextual attributes like user role, device, and time of access.
- **Machine Learning and Anomaly Detection:** Developing effective behavioral anomaly detection models and user behaviour profiles demand proficiency in supervised and unsupervised learning as well as real-time classification Techniques. This knowledge and expertise is essential to detect unauthorized activity under legitimate user sessions and during the authentication. Therefore, expertise in this domain makes the “SafeLAN” system success.
- **Blockchain Technology:** The implementation of immutable audit logs requires familiarity with permissioned blockchain platforms such as Hyperledger Fabric. This includes understanding smart contracts, consensus algorithms, and secure node configuration within a LAN.
- **Desktop Application Development:** Skills in developing secure, user-friendly desktop applications using frameworks like PyQt5 are vital for integrating GUI elements with backend systems, including encryption, authentication, and logging modules.

Data Requirements:

- The project requires datasets of user interaction behaviors, such as keystroke timing, mouse movements, to train biometric and anomaly detection components.
- Additionally, simulated or controlled access logs reflecting both normal and anomalous behavior will be used to develop and evaluate the blockchain-based logging and ML models effectively.

8. Objectives and Novelty

Main Objective Develop a secure file sharing, viewing, and storage file system for LAN environments to protect confidentiality and integrity.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
U H A P G Waidyarathne IT22328298	To implement a dynamic access control mechanism for LAN file system using behavioural biometrics.	- Collect and analyze keystroke and mouse movement data. - Calculate trust scores based on behaviour profile and contextual attributes like user role, device, and time of access. - Grant or deny access based on real-time scoring	Bridges the security gap in LAN-based access control systems by introducing a real-time biometric authentication layer, reducing the attack surface associated with relying solely on static passwords.
Chamindu Dilneth Wahalathanthri IT22029386	Behavior anomaly detection using machine learning to ensure continuous authentication	To evaluate and finalize the implementation of a machine learning-based behavioral anomaly detection system that ensures continuous authentication in LAN environments, specifically designed to detect and mitigate Physical Hijacking, Remote Session Hijacking,	This research introduces a multi-context, session-level continuous authentication system that leverages machine learning-based behavioral anomaly detection specifically to detect post-login hijacking attacks—an area that is often overlooked in

		and Man-in-the-Middle (MITM) Session Hijacking.	traditional access control systems.
Hirimuthugoda N A IT22323002	Secure file transmission and viewing method.	• Develop a secure file transfer protocol utilizing post-quantum cryptography. • Develop a secure file viewing system.	• Less prone to brute-force attacks than SSH. • Implement a biometric where the user must show the digits of the code received by hand.
Dias W P S IT22892744	To implement immutable and transparent audit logging using a permissioned blockchain for secure file access monitoring.	• Design and deploy a permissioned blockchain to store file access logs. • Record each access/modification event as a transaction with a timestamp and user ID. • Enable real-time log visibility through a secure browser interface for the owner and viewer. • Ensure logs are cryptographically linked and tamper-proof.	• Introduces real-time, dual-party log visibility via blockchain in a secure system, while ensuring transparency and non-repudiation. • Enhances forensic audit reliability by preventing log tampering and allowing shared accountability in LAN-based environments.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Waidyarathne U H A P G IT22328298	- This component directly aligns with the Cyber Security principle of Authentication and Access Control. By using a dynamic access control system based on real-time behavioural biometrics, the research strengthens the identity verification within LAN systems. - It reduces the need for static passwords that can be easily compromised when used solely and instead uses unique user behaviour patterns to authenticate access. This helps reduce risks from internal threats and illegal data access by making highly relevant for closed-network systems that need to be secure.
Chamindu Dilneth Wahalthanthri IT22029386	Project combines cybersecurity expertise (threats, authentication, intrusion detection) with machine learning techniques (behavior modeling, anomaly detection, algorithm development) making it highly relevant and compliant with a specialization in Cybersecurity, AI/ML, or Secure Software Systems.
Hirimuthugoda N A IT22323002	- Utilize post quantum cryptography to develop a more secure post file transfer protocol protecting confidentiality of it. - A secure file viewing system with an advanced biometric system that protects user's privacy rights.
Dias W P S IT22892744	- Records all file access events immutably using blockchain and cryptographic techniques to ensure integrity and non-repudiation. - Prevents tampering or deletion of logs, even by administrators, thereby reducing insider threats and supporting forensic investigations. - Promotes secure and transparent collaboration by allowing real-time dual-party visibility of audit logs through a secure system.

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 26

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Kanishka	Yapa	KPY
Co-Supervisor	Mr.	Uditha	Dharmakeerthi	Uditha 27/06
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

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Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.